DATA & AI CHALLENGE · INTELLIGENT CANDIDATE DISCOVERY & RANKING

Reading the profile, not the keywords.

A hybrid, CPU-only ranker that beats keyword-stuffers, surfaces plain-language talent, and avoids the honeypots.

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THE PROBLEM

Keyword filters can't see what matters

- **Recruiters miss the right person — not for lack of talent, but because keyword filters are blind to fit.**
  The JD is written to defeat keyword matching; the 'right answer' is reasoning about what the JD means.
- **The dataset is adversarial by design.**
  Keyword stuffers, plain-language strong candidates, behavioral twins, and ~80 'impossible' honeypots.
- **The shipped sample submission deliberately falls for it.**
  It ranks an HR Manager #1 and real ML Engineers at #27 / #48 / #99 — purely on AI-skill counts.
- **Our goal: a shortlist a recruiter can trust — and that survives code reproduction + interview.**
  NDCG@10 dominates scoring (0.50), so the top-10 must be genuinely excellent.

What the role actually needs

FIT (the ideal hire)

- 6–8 yrs, applied ML at product companies (not services)
- Shipped search / ranking / retrieval / recsys to real users
- Embeddings + vector/hybrid search + rigorous ranking eval
- Strong NLP/IR; in/near Pune–Noida; actually available

ANTI-FIT (explicit disqualifiers)

- Stuffed AI skills but a non-technical real title
- Pure research / academia with no production
- Only recent LangChain-on-OpenAI wrappers
- CV/speech-only · services-only · title-chasers · dormant

ARCHITECTURE

Four interpretable layers, no LLM at inference

```
score = (0.70·structured_fit + 0.30·semantic_fit) · behavioral_modifier · honeypot_gate
```

Structured fit
title coherence
career evidence
skill trust
domain · location

Semantic fit
contrastive
JD-ideal minus
JD-anti-pattern
static embeds

Behavioral
availability
recency · response
open-to-work
notice period

Honeypot gate
impossible
tenure / skills
→ removed from
top-100

Two-pass design: scalar scoring over 100K, then full facts + grounded reasoning for the top 100. CPU-only, streams the pool.

LAYER 1 — STRUCTURED FIT

The JD's logic, made machine-usable (70% weight)

- **Title coherence — the decisive anti-stuffer signal.**
  A 'Marketing Manager' with 9 AI skills scores ~0.06; bullseye AI/search/recsys titles score 1.0.
- **Career evidence — read from descriptions, not skill tags.**
  Credits demonstrated retrieval / ranking / production / NDCG-MAP evaluation / NLP-IR work.
- **Skill trust — defeats stuffing.**
  Skills reweighted by endorsements × duration × proficiency × platform assessment (raw lists ignored).
- **Plus: experience band (6–8), product-vs-services, NLP/IR-vs-CV/speech, location; disqualifier penalties.**
  Research-only, LangChain-only, and title-chaser flags apply explicit multiplicative penalties.

LAYER 2 — SEMANTIC FIT

Contrastive static embeddings (30% weight, CPU, no GPU)

- **sem = cos(candidate, JD-ideal) − cos(candidate, JD-anti-pattern).**
  Subtracting similarity to the unwanted profile is what mutes stuffers (close to both anchors).
- **Surfaces plain-language Tier-5s.**
  Someone titled 'Software Engineer' who describes building a recommendation system rises on semantics.
- **model2vec static embeddings — distilled for retrieval.**
  No transformer forward pass, no torch, no GPU; the latency/quality tradeoff the JD explicitly asks about.
- **Pure scikit-learn LSA fallback keeps it reproducible with zero downloads.**

Down-weight the unavailable; remove the impossible

Behavioral modifier

- 'Dormant 6 months + 5% response = not actually available.'
- Bounded multiplier from 23 signals, calibrated to the pool.
- Recency, response rate, open-to-work, notice, demand.

Honeypot gate

- Detects internal impossibilities, not specific IDs.
- Tenure > time elapsed; ≥3 expert skills at 0 months.
- 0 honeypots in our top-100 (cap is 10%).

WHAT THE DATA TOLD US

EDA on the full 100K pool drove every threshold

- **Skills are ~uniform random noise.**
  Every common skill sits at ~12% of profiles — 'semantic search' as a listed skill means almost nothing.
- **Only ~1,200 of 100,000 have a genuine AI/ML title.**
  ~68K carry the 12 non-technical 'stuffer' titles; the bullseye titles number in the dozens.
- **Honeypots are rare and detectable.**
  We gate 43 internally-impossible profiles pool-wide via tenure/skill contradictions.
- **Availability is a gradient, not a cliff.**
  Everyone is ≥45 days inactive (median 130) — recency is scaled to the real distribution.

RESULTS

A top-100 a recruiter can trust

95%	85%	6.4	0
based in India	open to work	mean years exp.	honeypots in top-100

Sample top-ranked reasoning (generated from the candidate's own facts, no hallucination):

#1 Senior AI Engineer, 5.9 yrs; career history shows building retrieval/search & ranking/recommendation systems in production with NDCG/MAP-style evaluation; at a product company; trusted skills: TensorFlow, Recommendation Systems, NLP; 30-day notice, open to work, based in Trivandrum.

#2 Senior Machine Learning Engineer (7.2 yrs); career history shows building retrieval/search & ranking/recommendation systems in production with NDCG/MAP-style evaluation; at a product company; trusted skills: scikit-learn, Recommendation Systems, QLoRA; 15-day notice, open to work, based in Noida.

#3 Lead AI Engineer with 6.7 yrs; career history shows building retrieval/search & ranking/recommendation systems in production with NDCG/MAP-style evaluation; at a product company; trusted skills: Vector Search, Recommendation Systems, Qdrant; 30-day notice, open to work, based in Jaipur.

#4 Senior NLP Engineer, 8.9 yrs; career history shows building retrieval/search & ranking/recommendation systems in production with NDCG/MAP-style evaluation; at a product company; trusted skills: Pinecone, PEFT, Elasticsearch; 30-day notice, open to work, based in Coimbatore.

REPRODUCIBLE & DEFENSIBLE

Built for Stage-3 reproduction and the interview

- **One command, CPU-only, no network at ranking time.**
  python rank.py --candidates candidates.jsonl --out submission.csv
- **Every score is inspectable.**
  Interpretable sub-scores per candidate — easy to defend in the 'walk through your architecture' interview.
- **No LLM at inference → scales to a 200K pool in production.**
  Exactly the latency/quality thinking the JD asks a Senior AI Engineer to demonstrate.
- **Honest, varied, fact-grounded reasoning passes manual review.**
  Specific facts + JD connection + acknowledged concerns + rank-consistent tone.